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**Paper Code : BS-M201/BSM201 Mathematics - IIA**

**UPID : 002005**

*Time Allotted : 3 Hours* *Full Marks :70*

*The Figures in the margin indicate full marks.*

*Candidate are required to give their answers in their own words as far as practicable*

## Group-A (Very Short Answer Type Question)

[ 1 x 10 = 10 ]

1. Answer *any ten* of the following :

(i) If the mean of the variate x and y are respectively 5 and 70 and the regression coefficient x on y is 0.08 then find regression line of x on y.

(ii) 6.If $r_{xy} = -2/3$ , u=2x+5, v=-2y+3, then find $r_{uv}$ .

(iii) A population is normally distributed with the following parameter: mean = 5, s.d.= 0.1. Find the mean and s.d. of the sample mean of size 25.

(iv) What is type I error?

(v) Probability of a student getting 1st class 2nd class and 3rd class at examination are 1/10, 2/5, 1/5 respectively. Find the probability that he fails.

(vi) Write two properties of a bivariate distribution.

(vii) If X has variance 9, Y has variance 5 and if X, Y are independent then find Var(2X+Y-5).

(viii) Find the variance of first n natural number.

(ix) Mean, mode and standard deviation of a distribution are 41, 45 and 8 respectively. Then find skewness.

(x) What is null and alternative hypothesis?

(xi) For a random variable X, $E\{(X-1)^2\}=10$ and $E\{(X-2)^2\}=6$ , then find E(X).

(xii) If X is a normal variate with mean 4 and s.d. 0.5 then P(3.8<x<4.3)=P(____<z<____).

## Group-B (Short Answer Type Question)

Answer *any three* of the following : [ 5 x 3 = 15 ]

2. A box contains 5 defective and 10 non defective lamps. 8 lamps are drawn in succession randomly without replacement. What is the probability that the 8th lamp will be the 5th defective. [5]

3. If X and Y have same s.d. then prove that X+Y and X-Y are uncorrelated. [5]

4. A coin is tossed 900 times and heads appear 490 times. Does the result support the hypothesis that the coin is unbiased? [5]
[ Given P(0<z<2.58)=0.495 ]

5. The following data relates to results of a fertiliser experiment on crops yields: [5]

| Fertiliser: | 0 | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|
| Units of fields : | 110 | 113 | 118 | 119 | 120 | 118 |

Fit a straight line to the above data and estimate the amount of yield when units of fertiliser used are 3 and 7 respectively.

6. Find the first two central moments for the variable x having following frequency distribution: [5]

| X: | 21 - 24 | 25 – 28 | 29 – 32 | 33 – 36 | 37 – 40 | 41 - 44 |
|---|---|---|---|---|---|---|
| Frequency: | 40 | 90 | 190 | 110 | 50 | 20 |

## Group-C (Long Answer Type Question)

Answer *any three* of the following : [ 15 x 3 = 45 ]

7. Find the first second third and fourth central moments of the frequency distribution. Hence find the skewness and kurtosis of the distribution. [ 15 ]

| Class limits | 100-104.9 | 105-109.9 | 110-114.9 | 115-119.9 | 120-124.9 |
|---|---|---|---|---|---|
| Frequency | 7 | 13 | 25 | 25 | 30 |

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8. Fit a second degree parabola to the given data by method of least square: [ 15 ]

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| x: | 7.5 | 10 | 12.5 | 15 | 17.5 | 20 | 22.5 |
| y: | 1.9 | 4.5 | 10.1 | 17.6 | 27.8 | 40.8 | 56.9 |

9. (a) In a sample of 500 student drawn from the College students in West Bengal it is observed that the number of students reading In HS, 1st year 2nd year 3rd year are 200, 150, 100, 50 respectively. Will it support the hypothesis that in West Bengal number of students reading HS, 1st year, 2nd year, 3rd year are in proportion 3:2:2:1. Test at 5% level of significance. Given that 5% value of Chi Square for 3 and 4 degree of freedom is 7.815 and 7.416 respectively. [ 7 ]

(b) In an experiment of immunisation of cattle from tuberculosis the following results were obtained: [ 8 ]

| | Affected | Not-Affected |
|---|---|---|
| Inoculated | 12 | 25 |
| Not-inoculated | 16 | 6 |

Calculate Exceeding Celsius Joseph Chi Square and discuss the effect of vaccine in controlling susceptibility of tuberculosis.
[5% value of Chi Square for one degree of Freedom is 3.84]

10. (a) Probability distribution of blood types in India is given in the table [ 5 ]

Types: O A B AB
Probability: 0.3712 0.2288 0.3226 0.0774

A random sample of 10 is considered from Indian population. What is the probability that there will be 4 Os, 3 As, 2 Bs and 1 AB blood types, respectively.

(b) A car hire firm has two cars to hire on a daily basis with the average number of demands per day as 1.5. Calculate the number of days when neither car is used and when some demands are refused if demand is distributed as poisson distribution. given e-1.5=0.2231. [ 5 ]

(c) A and B play a game in which their respective chances of winning are in the ratio 3:2. Find A's chances of winning atleast 3 out of 5 games played. [ 5 ]
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11. The joint pdf of bivariate (X,Y) is [ 15 ]

$f(x, y)=e^{-(x+y)}$ ,x>=0,y>=0
= 0 elsewhere

Find a) marginal pdf of X and Y
b) P(X+Y<4)
c) P(X>1)
d) P(X<Y)
e) P(2<X+Y<4)
f) Show that X and Y independent.

*** END OF PAPER ***

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